Exercise

1. If f is a polynomial and its degree is greater than 2, then f' is not constant.
   $P : f$ is a polynomial
   $Q : deg(f) > 2$
   $R : f'$ is not constant

   $P \wedge Q \Rightarrow R$

2. The number x is positive but the number y is not positive.
   P : The number x is positive
   Q : The number y is not positive.

   $P \wedge Q$

3. If x is prime, then $\sqrt{x}$ is not a rational number.
   $P : x$ is prime
   $Q : \sqrt{x} \notin \mathbb{Q}$

   $P \Rightarrow Q$

4. For every prime number p there is another prime number q with $q > p$.

   P is the set of prime numbers, $\forall p \in P, \exists q \in P, q > p$

5. For every positive number ϵ , there is a positive number δ for which $|x - a| < \delta$ implies $|f(x) - f(a)| < \epsilon$

   $\forall \epsilon \in \mathbb{R}, \epsilon > 0, \exists \delta \in \mathbb{R}, \delta > 0, (|x - a| < \delta) \Rightarrow (|f(x) - f(a)| < \epsilon)$

6. For every positive number ϵ there is a positive number M for which $|f(x) - b| < \epsilon$, whenever $x > M$

   $\forall \epsilon \in \mathbb{R}, \epsilon > 0, \exists M \in \mathbb{R}, M > 0, (x > M) \Rightarrow (|f(x) - b| < \epsilon)$

7. There exists a real number a for which $a + x = x$ for every real number x.

   $\forall x \in \mathbb{R}, \exists a \in \mathbb{R}, a + x = x$

9. If x is a rational number and $x \neq 0$, then $\tan(x)$ is not a rational number.
   $P : x \in \mathbb{Q}$
   $Q : x \neq 0$
   $R : \tan(x) \in \mathbb{Q}$

   $P \wedge Q \Rightarrow (\sim Q)$

10. If $\sin(x) < 0$, then it is not the case that $0 \leq x \leq \pi$
    $P : \sin(x) < 0$
    $Q : 0 \leq x \leq \pi$

    $P \Rightarrow \sim Q$

11. There is a Providence that protects idiots, drunkards, children and the United States of America. (Otto von Bismarck)
    S is the set $\{idiots, drunkards, children, USA\}$
    $P(x) : x$ is a providence
    $Q(x, y) : x$ protects y

    $\forall y \in S, \exists x, P(x) \wedge Q(x, y)$

12. You can fool some of the people all of the time, and you can fool all of the people some of the time, but you can't fool all of the people all of the time. (Abraham Lincoln)
    T is the set of the all times, P the set of all the people
    $Q(x, y) =$ Fooling x people for time y

    $((\exists x \in P, \forall y \in T, Q(x, y)) \vee (\forall x \in P, \exists y \in T, Q(x, y)) \wedge \sim (\forall x \in P, \forall y \in T, P(x, y))$
    $= (\exists x \in P, \forall y \in T, Q(x, y)) \vee (\forall x \in P, \exists y \in T, Q(x, y)) \wedge (\exists x \in P, \exists y \in P, \sim Q(x, y))$

13. Everything is funny as long as it is happening to somebody else. (Will Rogers)
    X is the set of everything
    $P : x$ is funny
    $Q : x$ is happening to you
    $R : x$ is happening to someone

    $\forall x \in X, (\sim Q \wedge R \Rightarrow P)$